Akkshay Sundara Rajan

+1 (765) 772-5548 | akkshaysr0107@gmail.com | github.com/akkshay0107 | linkedin.com/in/akkshaysr

Experience

Project Lead Sep 2025 — Present
Machine Learning @ Purdue - Student Organization *West Lafayette, Indiana*

- Lead 6 developers in building a BERT style transformer policy network for text-state modeling and PPO training in a partially observed environment, achieving an 86% win rate against benchmark heuristics.
- Improved policy robustness and diversity through a league-based training pipeline inspired by AlphaStar and OpenAI Five.
- Architected an asynchronous WebSocket worker queue for distributed inference and simulation, reducing action latency to 25 ms and increasing rollout throughput by 60% through output caching.
- Deployed a CPU optimized model on Google Compute Engine to support 20+ concurrent battles per node.

Undergraduate Researcher / ML Engineer Aug 2025 — Dec 2025
Dr Philip Low's Lab *West Lafayette, Indiana*

- Built a distributed protein structure inference pipeline using NVIDIA Boltz-2, scaling and managing prediction workloads across 8 HPC nodes through SLURM scripts.
- Developed a postprocessing workflow (OpenMMDL) to test the stability of predicted complexes for drug discovery research.

Software Engineering Intern Jun 2025 — Aug 2025
Siemens Energy *Gurgaon, India*

- Developed an ETL pipeline using Python and SQL on Azure to manage and analyze 10 million sensor data records.
- Implemented an early-warning system using decision trees in scikit-learn to detect high outlet temperatures in gas turbines.
- Created a FFT-based diagnosis system to classify turbine failure modes by matching sensor data against past failures.
- Automated the visualization of multi sensor feeds in a dashboard using plotly, dash and Flask saving 1 hour per inspection.

Undergraduate Researcher Feb 2025 — Sep 2025
AIDA^3 Labs, Purdue *West Lafayette, Indiana*

- Prototyped a medical emergency response agent using Llama 3 and LangChain to support dispatch coordination.
- Designed human-in-the-loop route planning workflows using Google Maps MCP data with LLM reasoning over call logs.

Education

Purdue University West Lafayette, IN
Bachelor of Science, Computer Science and Mathematics (Double Major) *Aug 2024 — (Expected) Dec 2027*

- Dean's List (4x), Semester Honors (4x), 3.92 GPA
- Coursework: Data Mining, Machine Learning, Data Structures and Algorithms, Linear Algebra, Statistics, DevOps Tools

Projects

sift - Multimodal Semantic Search Engine github.com/akkshay0107/sift

- Engineered a high-performance semantic search engine for multimodal retrieval across text, image, audio, and video files.
- Built multimodal ingestion pipelines with OCR, audio transcription (Whisper), and Qwen3-VL to embed files into a local Qdrant vector database, optimized to minimize RAM consumption by 40% using BF16 quantization.
- Architected a real-time, event-driven indexing daemon using watchdog and content hashing for zero redundancy updates.

glimpse - AI Newsletter Automation Platform github.com/sanjayriram44/glimpse

- Built a full-stack newsletter SaaS with FastAPI, Next.js, and Supabase that consolidates email subscriptions into a single AI-generated daily digest in text and audio using Gemini and ElevenLabs TTS.
- Designed an async Celery/Redis email ingestion system with Gmail OAuth, for scalable background processing.
- Automated daily summarization via LangChain deployed as a cron job, with storage using AWS S3 buckets.

Skills

- **Programming Languages**: Python, C++, Rust, Java
- **ML/Data**: PyTorch, Langchain, Scikit-learn, HuggingFace, Pandas, Qdrant, Numpy, SQL, Postgres
- **Backend/Cloud**: FastAPI, Docker, Redis, Celery, Azure, AWS, Google Cloud Platform, SLURM, Bash